C-STGB: Real-Time Financial Surveillance Architecture & Conformal Triage Platform

(a) Streaming Ingestion & Deep Spatio-Temporal GraphBoost Neural Core

Fast-Path Latency: 0.45 ms | Throughput: 31,200 tx/s

Stage 1: Streaming Ingest

SWIFT MT103 | ISO-20022 | Kafka

UTXO Blockchains (Bitcoin, Eth)

DuckDB Streaming Micro-Batches

Apache Arrow Zero-Copy Buffer

Causal Split: $t_e \le t_{\text{eval}}$ **(No Leak)**

Ingestion Latency SLA: 0.11 ms

Stage 2: 12-D Invariants

Mass Flow: $\Phi_{\text{flow}} = \ln(1 + \frac{A_{\text{out}}}{A_{\text{in}} + \varepsilon})$

Pass-Through Layering Detector

Hawkes: $\lambda_u(t) = \mu_u + \sum \alpha e^{-\beta \Delta t}$

Burst Acceleration (μ_A, σ_A^2)

Directed Taint: $s_{\text{fwd}}, s_{\text{bwd}}$

Tensor: $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$

Stage 3: Temporal GNN

3-Band Harmonic Attention

$\tau_1 = 2.4\,\text{h}, \tau_2 = 24\,\text{h}, \tau_3 = 168\,\text{h}$

Edge Gate: $\hat{g}_{ij} \ge \delta_{\text{floor}}$ **(Prunes 65.9%)**

Adversarial Chaff Link Suppression

Typology Latent SMOTE on $\mathcal{C}_k$

GNN Forward SLA: 0.16 ms

Stage 4: Fusion & Risk

Evidence Gate $\alpha_u \in [0, 1]$

$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$

Cold-Start: 89.60% F1 for $\deg(u) \le 2$

Cost-Sensitive Head: $\tau^* = 0.18$

15:1 Asymmetric Ratio $(C_{\text{FN}}/C_{\text{FP}})$

Risk Output: $\hat{p}_u \in [0, 1]$

(b) Class-Conditional Conformal Risk Control (CRC) Triage Gate & Operational Workflow Clearing

Operational Coverage: 99.77% | Empty Sets: 0.00%

Holdout Calibration

Holdout Split: $\mathcal{D}_{\text{cal}}$

$n_0 = 3,306,\ n_1 = 372$

Non-Conformity Scores:

$s_i^{(1)} = 1 - \hat{p}_i$

$s_i^{(0)} = \hat{p}_i$

Target Coverage: 99.0%

Class-Cond. CRC Triage

$1 - \alpha \ge 99\%$

$\hat{q}^{(0)}, \hat{q}^{(1)}$

Tier 1: High-Risk Quarantine $[\Gamma = \{1\}]$

Rate: 18 / 10,000 tx (0.18%)

Immediate Quarantine Hold | Automated Wire Freeze

FinCEN Form 111 SAR Dossier Draft | GNNExplainer Attribution | SHA-256 Merkle Receipt

Tier 2: Compliance Review Desk $[\Gamma = \{0, 1\}]$

Rate: 49 / 10,000 tx (0.49%) | Queue Cut: -99.51%

Human-in-the-Loop 4-Eyes Compliance Review Queue

Ambiguous boundary triage routed to senior examiners | 85.7%--95.8% fewer false alerts vs rules

Tier 3: Straight-Through Automated Clear $[\Gamma = \{0\}]$

Rate: 9,933 / 10,000 tx (99.33%)

Automated Wire-Speed Straight-Through Processing (STP)

Wire-speed release under 0.45 ms SLA | Mathematically bounded selective risk (< 0.13% residual FN)

REAL-TIME PRODUCTION SLAS | STATISTICAL SAFETY GUARANTEES | REGULATORY GOVERNANCE

Latency Performance
0.45 ms Fast-Path
P99: 0.82 ms | 31,200 tx/s
Strict compliance with <10 ms SLA

Workload Efficiency
-99.51% Review Cut
Only 49 of 10,000 tx reviewed
85.7%--95.8% fewer false alerts

Statistical Safety
$\ge 99.0\%$ **Coverage**
Finite-sample exchangeability bound
0.00% empty sets | Avg size: 1.005

Regulatory Governance
SR 26-2 & OCC Ready
Zero-hallucination graph evidence
SHA-256 Merkle cryptographic trail